

Changhun Kim

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Research Interests

My research centers on **deep generative models**, particularly **diffusion models**, spanning theory and algorithms: why they **generalize**, when they **hallucinate**, and how to make them more **trustworthy**. Building on these foundations, I extend this line of research to **embodied AI**, including **world models** and **vision-language-action models**.

Keywords: Trustworthy Machine Learning, Deep Generative Models, Embodied AI, Machine Learning Theory

Education

University of Wisconsin–Madison Madison, WI, USA
Ph.D. in Computer Science (Advisor: Prof. Grigorios Chrysos) Aug. 2026 – Present

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
M.S. in Artificial Intelligence (Advisor: Prof. Eunho Yang) Feb. 2022 – Feb. 2024
B.S. in Computer Science and Mathematics Feb. 2017 – Feb. 2022

Employment

UW–Madison Chrysos Lab Madison, WI, USA
Graduate Research Assistant (Advisor: Prof. Grigorios Chrysos) Aug. 2026 – Present

UBC Computer Vision Lab Vancouver, BC, Canada (Remote)
External Collaborator (Advisor: Prof. Evan Shelhamer) Jul. 2025 – Present

AITRICS Seoul, Korea
Researcher (Advisor: Prof. Eunho Yang, Prof. Juho Lee) Nov. 2023 – Jun. 2026

KAIST Machine Learning and Intelligence Lab Daejeon, Korea
Graduate Research Assistant (Advisor: Prof. Eunho Yang) Feb. 2022 – Feb. 2024
Undergraduate Research Assistant (Advisor: Prof. Eunho Yang) Jun. 2021 – Feb. 2022

KAIST Vehicular Intelligence Lab Daejeon, Korea
Undergraduate Research Assistant (Advisor: Prof. Dongsoo Har) Oct. 2019 – Aug. 2020

Publications

Manuscripts *: Equal Contribution, †: Equal Advising

[4] **ChronoFlow: Hierarchical Flow Matching for Irregular Time Series Generation**
Changhun Kim*, Sunguk Jang*, Jeongjun Lee, Juhwan Choi, Sangchul Hahn, Grigorios Chrysos, Eunho Yang†, Juho Lee†
Under Review at International Conference on Learning Representations (ICLR), 2027

[3] **Domain Generalization under Sampling Pattern Shifts in Irregularly Sampled Time Series**
Changhun Kim, Joohyung Lee, Kwanhyung Lee, Donghwee Yoon, Grigorios Chrysos, Eunho Yang
Under Review at International Conference on Learning Representations (ICLR), 2027

- [2] **TRIAGE: Dialectical LLM Reasoning for Explainable Risk Prediction on Irregularly Sampled Medical Time Series**
Hyeongwon Jang*, Gyouk Chu*, **Changhun Kim**, Hangyul Yoon, Jeonguk Lee, Eunho Yang, Joonhyung Park
Under Review at International Conference on Learning Representations (ICLR), 2027
- [1] **Development and External Validation of a Transformer-Based Cardiac Arrest Prediction System for General Ward Patients: A Multicenter Study**
Sungsoo Hong*, Kyung Hyun Lee*, Yechan Mun*, Sangchul Hahn, Sunguk Jang, **Changhun Kim**, Yuhyun Choi, Kyung Soo Chung, Kang-Mo Gu, Hae In Jung, Tae Wan Kim, Moon Seong Baek, Taeyong Sim
Under Review at BMC Medical Informatics and Decision Making, 2026

Conference Publications

- [8] **ReviewScore: Misinformed Peer Review Detection with Large Language Models**
Hyun Ryu, Doohyuk Jang, Hyemin S. Lee, Joonhyun Jeong, Gyeongman Kim, Donghyeon Cho, Gyouk Chu, Minyeong Hwang, Hyeongwon Jang, **Changhun Kim**, Haechan Kim, Joowon Kim, Yoonjeon Kim, Kwanhyung Lee, Chanjae Park, Heecheol Yun, Gregor Betz, Eunho Yang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026 (Findings)
- [7] **Delta-XAI: A Unified Framework for Explaining Prediction Changes in Online Time Series Monitoring**
Changhun Kim*, Yechan Mun*, Hyeongwon Jang, Eunseo Lee, Sangchul Hahn[†], Eunho Yang[†]
International Conference on Learning Representations (ICLR), 2026
- [6] **Soft Equivariance Regularization for Invariant Self-Supervised Learning**
Joohyung Lee, **Changhun Kim**, Hyunsu Kim, Kwanhyung Lee, Juho Lee
International Conference on Learning Representations (ICLR), 2026
- [5] **Position Paper: How Should We Responsibly Adopt LLMs in the Peer Review Process?**
Juhwan Choi, JungMin Yun, **Changhun Kim**, YoungBin Kim
Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026 (Findings)
- [4] **TIMING: Temporality-Aware Integrated Gradients for Time Series Explanation**
Hyeongwon Jang*, **Changhun Kim***, Eunho Yang
International Conference on Machine Learning (ICML), 2025 (Spotlight, 313/12,107=2.6%)
ICLR Workshop on XAI4Science, 2025
CIKM Workshop on Human-Centric AI, 2025 (Best Paper Award)
- [3] **Stable-TTS: Stable Speaker-Adaptive Text-to-Speech Synthesis via Prosody Prompting**
Wooseok Han*, Minki Kang*, **Changhun Kim**, Eunho Yang
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025
- [2] **CloudFixer: Test-Time Adaptation for 3D Point Clouds via Diffusion-Guided Geometric Transformation**
Hajin Shim*, **Changhun Kim***, Eunho Yang
European Conference on Computer Vision (ECCV), 2024
- [1] **SGEM: Test-Time Adaptation for Automatic Speech Recognition via Sequential-Level Generalized Entropy Minimization**
Changhun Kim, Joonhyung Park, Hajin Shim, Eunho Yang
Conference of the International Speech Communication Association (INTERSPEECH), 2023 (Oral)

Workshop Papers

- [3] **Structure-Aware Set Transformers: Temporal and Variable-Type Attention Biases for Asynchronous Clinical Time Series**
Joohyung Lee, Kwanhyung Lee, **Changhun Kim**, Eunho Yang
ICLR Workshop on Time Series in the Age of Large Models, 2026

- [2] **DeltaSHAP: Explaining Prediction Evolutions in Online Patient Monitoring with Shapley Values**
Changhun Kim*, Yechan Mun*, Sangchul Hahn, Eunho Yang
ICML Workshop on Actionable Interpretability, 2025
- [1] **AdapTable: Test-Time Adaptation for Tabular Data via Shift-Aware Uncertainty Calibrator and Label Distribution Handler**
Changhun Kim*, Taewon Kim*, Seungyeon Woo, June Yong Yang, Eunho Yang
NeurIPS Workshop on Table Representation Learning, 2024

Awards and Honors

Awards

Best Paper Award, CIKM 2025 Workshop on Human-Centric AI	Nov. 2025
Top Reviewer (206/10,943=1.9%), ICML 2025	May 2025
Silver Prize, Korean Undergraduate Mathematics Competition	Jan. 2022

Honors

Korean Government Scholarship Program for Study Overseas, National Institute for International Education (NIIED)	Fall 2026 – Present
Full Doctoral Assistantship, University of Wisconsin–Madison	Fall 2026 – Present
Dongwon Full Master’s Scholarship, Dongwon Group	Spring 2022 – Fall 2023
Magna Cum Laude, KAIST College of Engineering	Feb. 2022
National Full Undergraduate Scholarship, Korea Student Aid Foundation (KOSAF)	Spring 2017 – Fall 2021
KAIST Convergence AMP Scholarship, KAIST School of Computing	Mar. 2019

Teaching Experience

Teaching Assistant at UW–Madison

CS220 Data Science Programming I	Fall 2026
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Mentoring Experience

Hyeongwon Jang, Ph.D. Student at KAIST	Oct. 2024 – May 2026
Vishal Srivastava, Ph.D. Student at NUS	Aug. 2025 – Apr. 2026
Wooseok Han, Researcher at AITRICS	Mar. 2024 – Nov. 2024
Taewon Kim, Undergraduate Student at KAIST	Mar. 2023 – Nov. 2023
Sungwoo Cho, Undergraduate Student at KAIST	Sept. 2022 – Apr. 2023

Academic Service

Journal Reviewer: TMLR, TNNLS, Neurocomputing, IJCV

Conference Reviewer: NeurIPS, ICML, ICLR, AISTATS, AAAI, IJCAI, ACL, EMNLP, ICASSP, LoG

Invited Talks

Delta-XAI: A Unified Framework for Explaining Prediction Changes in Online Time Series Monitoring

IEIE Workshop on AI & Semiconductor Technologies for Future Healthcare	May 2026
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